P2P Shopping

Roluri:
Irina: UML

Iulia: Use Case diagram

Diana: Requirement list

AI tools used: Gemini
Given prompt: Act as a software engineer and draft a technical overview for P2P Shopping project, a shopping assistant that uses crowdsourced GPS/Wi-F**i** data to map supermarket layouts. Provide a concise project description, define the system actors (Shopper, Collaborative Group, Map Engine, Cloud Repository, and Positioning Provider), and outline use case scenarios for list management, P2P mapping, and route optimization. Ensure the response is brief, professional, and uses both text paragraphs and correct, specific bullet points.

Description:

CrowdShop P2P is an intelligent shopping assistant that transforms standard lists into a dynamic, crowdsourced navigation system. By leveraging GPS and Wi-Fi RTT, the app tracks where users mark items as purchased to build a real-time, user-generated map of supermarket layouts.

Features

- Crowdsourced Mapping: Automatically maps product locations based on aggregated user data.
- Route Optimization: Calculates the fastest path through a store to find every item on your list efficiently.
- Smart Store Finder: Recommends the best local supermarket based on where your specific items were recently spotted.
- Real-Time Collaboration: Syncs shared lists instantly, allowing multiple users to divide a single shopping trip.
- Advanced Planning: Full support for recurring items, price tracking, and detailed product metadata.

Based on this crowdsourced data, the application can guide users to products on their list, generate an efficient shopping route, suggest suitable supermarkets, and support real-time shared shopping lists.

Actors:

Individual Shopper:
The primary user of the application who creates and manages personal shopping lists, adds optional metadata(prices and brands) and marks items as completed while navigating through the store.

Collaborative List Owner (Organizer of a collaborative list): The user who creates the shared shopping list and manages it by inviting other participants, assigning permissions, and defining rules such as who is allowed to add, edit, or complete items.

Co-Shopper: A user who participates in a shared shopping list, being able to add items and mark them as completed in real time together with other users.

Location Service: An external system that provides the user's location (GPS or Wi-Fi) so the application can record where items are found in the store.

Inventory Data Provider (supermarket): A potential API from retail chains that could provide official product availability to supplement the user-generated data.

Cloud Database: Stores and synchronizes all application data, including supermarket layouts (aisle maps), the catalog of stores, and users' shopping lists.

System Administrator: The person responsible for managing and maintaining the system, ensuring proper operation, monitoring data integrity, and handling technical issues.

Use Case scenarios:

1. A user creates a new shopping list.

2. A user can add, edit, and delete items from the list.

3. For each item, the user can optionally add metadata (such as brand or price).

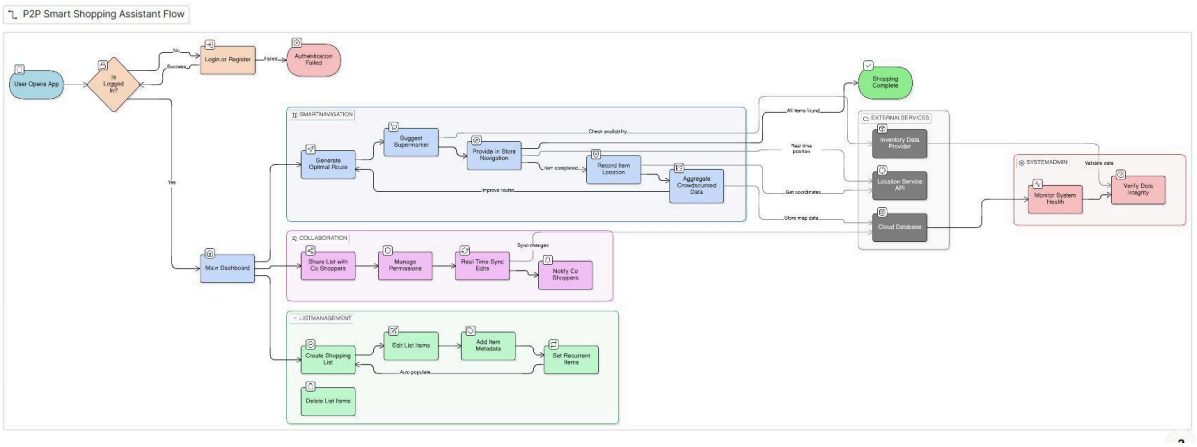
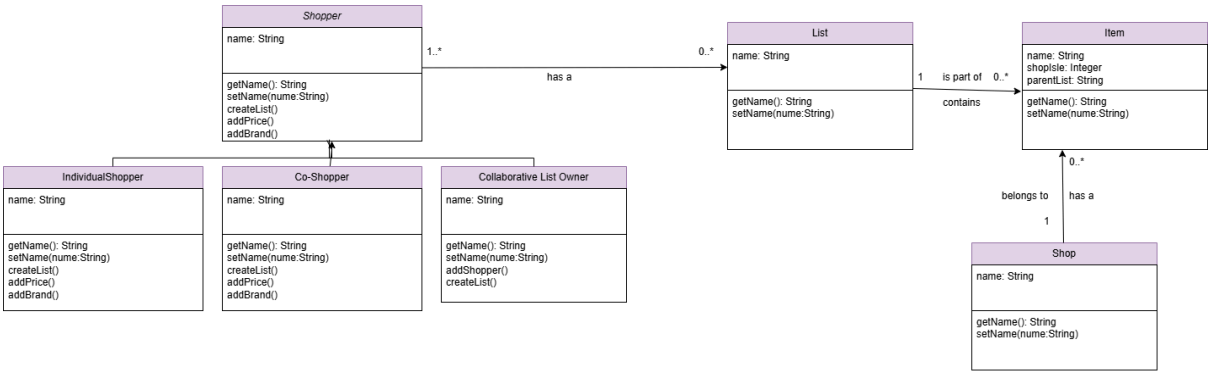
4. The list owner can share the list with co-shoppers.

5. The list owner can remove co-shoppers from the shared list.

6. The list owner can assign specific permissions to co-shoppers.

7. The list owner and the co-shoppers can edit the list and mark products as completed simultaneously.

8. The user selects a product from the list and is guided to the corresponding aisle.

9. The system administrator monitors data.

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